S16 — Do location-typical cue words impair letter identification at that location? (error rates)

Question. After reading a word for an object with a typical location (e.g., *hat*: up; *boot*: down), do people make more errors identifying a briefly shown letter (X/O) when it appears at the word's typical location than at the opposite location?

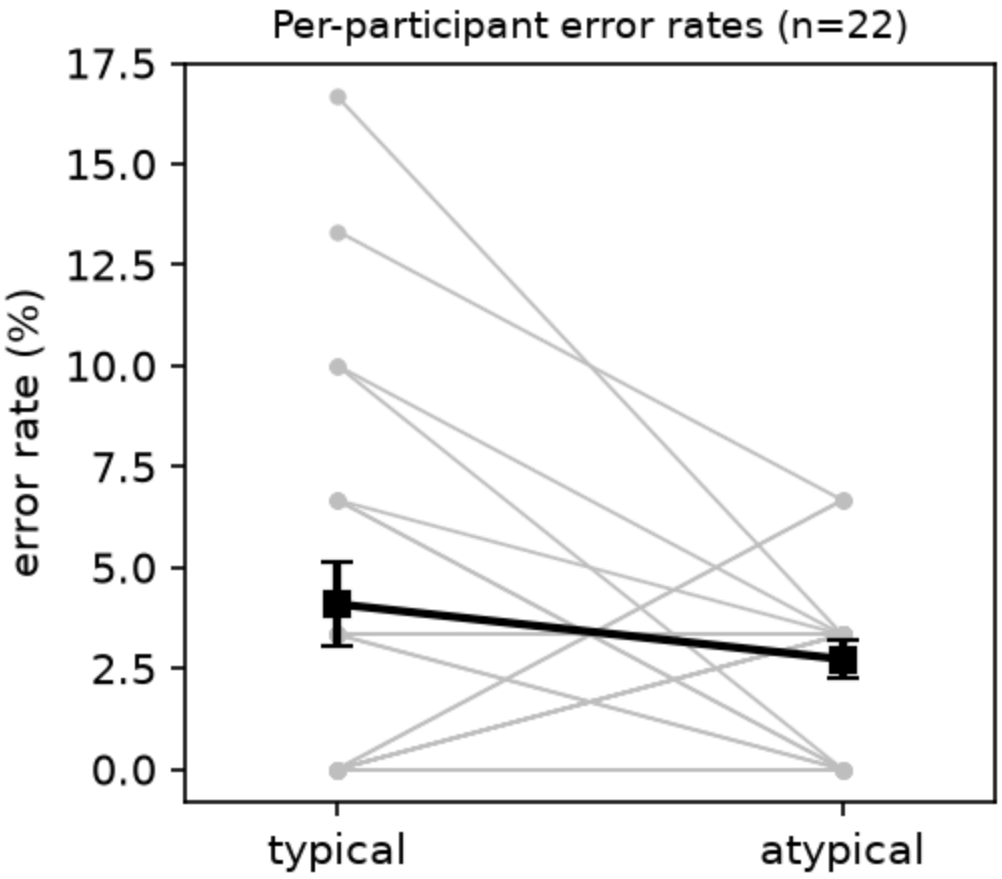
Data and sample. Replication data (`d02.sav`): 22 participants × 60 experimental trials = 1,320 trials, 30 typical and 30 atypical trials per participant. No participants or trials excluded (as the card prescribes for error rates; no missing values; `resp_corr` agrees with the `Correct` string on 100% of trials). Total errors: 45 (27 typical, 18 atypical).

Test (fixed in the log before analysis). Per-participant error rate = proportion of trials with `resp_corr` = 0, per condition (`typisch` 1 = typical, 2 = atypical). One-way repeated-measures ANOVA by subjects (F1), typical vs atypical; partial η^2 = SS_location / (SS_location + SS_error), 95% CI by noncentral-F inversion. The claim holds if typical > atypical with $p < .05$.

Result.

	Typical	**Atypical**
Mean error rate (SD)	4.09% (4.82)	2.73% (2.21)

- **$F(1, 21) = 1.45$, $p = .242$, partial $\eta^2 = .065$, 95% CI [.00, .31].**
- Mean difference (typical – atypical) = 1.36 percentage points, 95% CI [−0.99, 3.72]; d_z = 0.26.
- 9 participants made more errors at the typical location, 7 fewer, 6 tied.
- Robustness (not primary): Wilcoxon signed-rank W = 44.5, p = .216; trial-level logistic GEE clustered by participant, OR = 1.52, p = .176. Same direction, also not significant.



Interpretation. The difference is in the predicted direction but small and not distinguishable from zero. Errors are rare (3.4% overall, at most 5 of 30 trials per cell), so the test has little resolution: the interval for the difference runs from about a 1-point advantage for typical to a 3.7-point disadvantage, and the η^2 interval includes both zero and sizeable effects. The data therefore do not support the claim, without ruling out a modest effect.

Conclusion. The claim does not hold on this data: participants did not make significantly more identification errors at the cue word's typical location than at the atypical location ($F(1, 21) = 1.45$, $p = .24$).